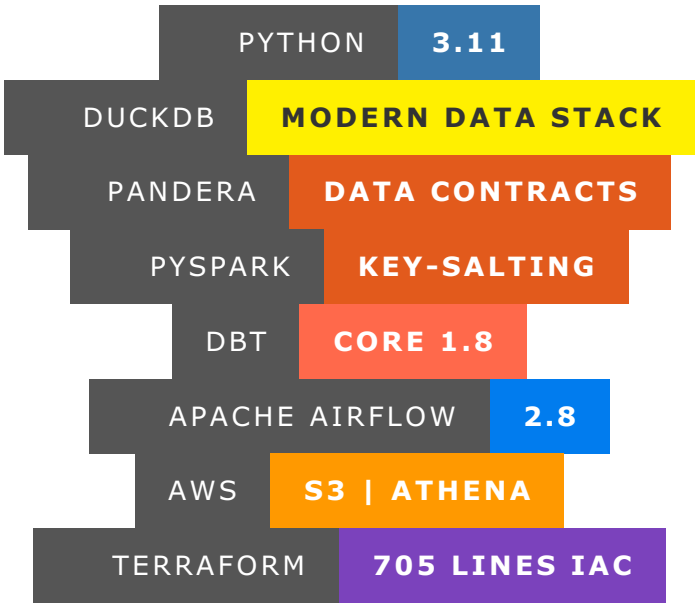
Coverdrive: Shift-Left Data Quality & Cricket Analytics Lakehouse (coverdrive)



An end-to-end analytical data lakehouse processing complex match datasets from ESPN Cricinfo and Cricsheet across **5,591+ T20 matches** and **1,264,534+ ball-by-ball delivery records**. The project introduces a "shift-left" data quality paradigm, enforcing strict Pandera data validation contracts directly inside Airflow orchestration tasks prior to warehouse ingestion. High-cardinality analytical joins are optimized across PySpark compute engines using key-salting skew reduction algorithms before loading into DuckDB.

System Performance Highlights

Metric	Target / Benchmark	Operational Result
Ingested Match Telemetry Volume	{{content}}gt;1,000,000$ Deliveries	**1,264,534 Ball-by-Ball Records** across 5,591 T20 Matches
Lakehouse Storage Scale	Zero-Maintenance S3	**461.6 MB across 9,813 Parquet Objects** in AWS S3
PyTest Test Coverage Suite	{{content}}gt;60%$ Coverage	**68.08% Automated Coverage** (35/35 Passing Unit/Integration Tests)
dbt Data Quality Assertions	$100%$ Contract Passing	**42/42 dbt Test Models Passed** (Zero schema drift violations)
Athena Serverless Analytics Latency	{{content}}lt;1.0\text{s}$ Response	**515 ms Response** (Scanning 4.11 KB on S3 Parquet)
Infrastructure Rigor	Declarative IaC	**705 Lines of Verified AWS Infrastructure-as-Code**

EXECUTIVE SUMMARY & BUSINESS PROBLEM

Analytical data platforms frequently suffer from data drift: source API schema changes silently corrupt downstream dashboards, forcing engineers into reactive debugging. In sports analytics processing millions of ball-by-ball events, data quality errors directly distort player performance models.

coverdrive addresses this challenge by shifting data quality left. Runtime schema contracts validate incoming match data at the Airflow DAG task level before it enters the analytical engine. Invalid data drops early into quarantine stores, protecting downstream DuckDB analytical models from structural corruptions.

ENTERPRISE AWS CLOUD PRODUCTION ARCHITECTURE

Cost Optimization Architecture Note: To maintain strict development cost efficiency ($0 cloud spend during iteration), this repository includes a dual-target execution engine:

1. **Local Development Target:** Emulates AWS S3 via MinIO, runs local Dockerized Airflow, and executes dbt transformations via dbt-duckdb.
2. **Production AWS Cloud Target:** Provisioned via 705 lines of Terraform IaC (infra/terraform/main.tf), executing dbt model transformations via **dbt-athena** over **AWS Glue Data Catalog**, backed by **EMR Serverless Spark**, **ElastiCache Redis**, and **FastAPI on ECS Fargate**.

```
flowchart LR
    %% Modern Professional Color Palette Definitions (ByteByteGo / IEEE Publication Style)
    classDef card_python fill:#FFFFFF,stroke:#3B82F6,stroke-width:2px,color:#0F172A;
    classDef card_s3 fill:#FFFFFF,stroke:#F59E0B,stroke-width:2px,color:#0F172A;
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    classDef card_spark fill:#FFFFFF,stroke:#EA580C,stroke-width:2px,color:#0F172A;
    classDef card_dbt fill:#FFFFFF,stroke:#EF4444,stroke-width:2px,color:#0F172A;
    classDef card_athena fill:#FFFFFF,stroke:#0284C7,stroke-width:3px,color:#0F172A;
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    classDef card_pandera fill:#FFFFFF,stroke:#A855F7,stroke-width:2px,color:#0F172A;
    classDef card_glue fill:#FFFFFF,stroke:#8B5CF6,stroke-width:2px,color:#0F172A;
    classDef card_dlq fill:#FEF2F2,stroke:#EF4444,stroke-width:2px,color:#991B1B;

    subgraph VPC ["     AWS PRIVATE VPC (SECURITY & IAM PERIMETER)"]
        direction LR

        %% STAGE 1: INGESTION (3 Multi-Source Ingestion Feeds)
        subgraph S1 ["STEP 1: INGESTION"]
            direction TB
            Src1["     <b>Cricsheet Data</b><br/><span style='color:#64748B;'>Match JSON / CSV</span>"]:::card_python
            Src2["🌐 <b>ESPNcricinfo</b><br/><span style='color:#64748B;'>HTML Scrapers</span>"]:::card_python
            Src3["     <b>Open-Meteo API</b><br/><span style='color:#64748B;'>Weather REST Feeds</span>"]:::card_python
            Ingest["<img src='https://raw.githubusercontent.com/devicons/devicon/master/icons/python/python-original.svg' width='30'/><br/><b style='color:#2563EB;'>AWS ECS Fargate</b><br/><span style='color:#64748B;'>Python Ingest Tasks</span>"]:::card_python
            Src1 --> Ingest
            Src2 --> Ingest
            Src3 --> Ingest
        end

        %% STAGE 2: BRONZE & QUALITY
        subgraph S2 ["STEP 2: BRONZE & QUALITY"]
            direction TB
            Bronze["<img src='https://raw.githubusercontent.com/devicons/devicon/master/icons/amazonwebservices/amazonwebservices-original-wordmark.svg' width='30'/><br/><b style='color:#D97706;'>AWS S3 Bronze Lake</b><br/><span style='color:#64748B;'>Raw Parquet Storage</span>"]:::card_s3
            Airflow["<img src='https://raw.githubusercontent.com/devicons/devicon/master/icons/apacheairflow/apacheairflow-original.svg' width='30'/><br/><b style='color:#017CEE;'>AWS MWAA Airflow</b><br/><span style='color:#64748B;'>Managed Orchestrator</span>"]:::card_airflow
```

```
        Pandera["    <br/><b style='color:#C026D3;'>Pandera Contracts</b><br/><span
style='color:#64748B;'>Shift-Left Schema Guard</span>"]:::card_pandera
        DLQ["    <br/><b style='color:#DC2626;'>S3 DLQ Bucket</b><br/><span
style='color:#991B1B;'>Quarantine Bad Data</span>"]:::card_dlq

        Airflow --> Pandera
        Pandera -.->|Invalid Schema| DLQ
    end

    %% STAGE 3: COMPUTE & GOVERNANCE
    subgraph S3 ["STEP 3: COMPUTE & GOVERNANCE"]
        direction TB
        Spark["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/apachespark/apachespark-
original.svg' width='30'/><br/><b style='color:#EA580C;'>EMR Serverless Spark</b><br/><span
style='color:#64748B;'>AQE Skew-Join Processing</span>"]:::card_spark
        Glue["    <b>AWS Glue Catalog</b><br/><span style='color:#64748B;'>Centralized
Metastore</span>"]:::card_glue
        Silver["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/amazonwebservices/amazon
webservices-original-wordmark.svg' width='30'/><br/><b style='color:#D97706;'>AWS S3 Silver
Lake</b><br/><span style='color:#64748B;'>Cleaned & Typed Parquet</span>"]:::card_s3
        dbt["    <br/><b style='color:#EF4444;'>dbt-Athena Core 1.8</b><br/><span
style='color:#64748B;'>42/42 Passed Quality Tests</span>"]:::card_dbt
        Gold["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/amazonwebservices/amazon
webservices-original-wordmark.svg' width='30'/><br/><b style='color:#D97706;'>AWS S3 Gold
Lake</b><br/><span style='color:#64748B;'>Enriched Player Marts</span>"]:::card_s3

        Spark --> Silver
        Silver --> dbt
        dbt --> Gold
        Glue -. Metastore Sync .-> Silver
        Glue -. Metastore Sync .-> Gold
    end

    %% STAGE 4: SERVING & ANALYTICS
    subgraph S4 ["STEP 4: SERVING & ANALYTICS"]
        direction TB
        Athena["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/amazonwebservices/amazon
webservices-original-wordmark.svg' width='30'/><br/><b style='color:#0284C7;'>AWS Athena
SQL</b><br/><span style='color:#64748B;'>Serverless Query Engine</span>"]:::card_athena
        Redis["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/redis/redis-original.svg'
width='30'/><br/><b style='color:#DC2626;'>ElastiCache Redis</b><br/><span
style='color:#64748B;'>Sub-10ms Hot Query Cache</span>"]:::card_redis
        FastAPI["<img
src='https://raw.githubusercontent.com/devicons/devicon/master/icons/fastapi/fastapi-original.svg'
width='30'/><br/><b style='color:#10B981;'>FastAPI on ECS</b><br/><span
style='color:#64748B;'>Stateless REST API</span>"]:::card_fastapi

        Athena --> Redis
        Redis --> FastAPI
    end

    %% SEQUENTIAL DATA FLOW CONNECTIONS
    Ingest -->|1. Write Raw| Bronze
    Bronze -->|2. Validate| Pandera
    Pandera -->|3. Clean & Process| Spark
```

```
    Gold -->|4. Query Gold| Athena
end
```

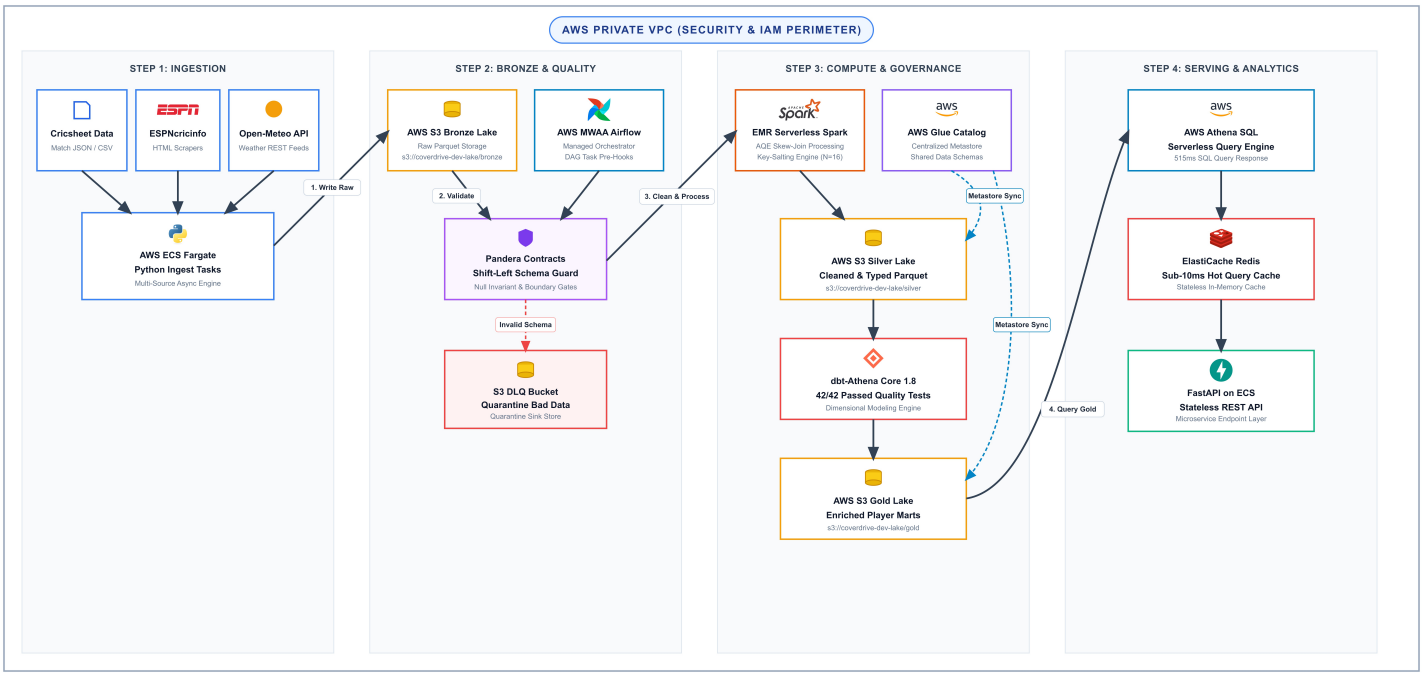

Figure 1: High-Resolution Pure AWS Cloud Data Pipeline Architecture Diagram.

Two-Minute Enterprise Architecture Defense Strategy

> *"When building Coverdrive, the principal architectural objective was eliminating downstream analytical corruption through shift-left data contracts and resolving executor memory skew across high-cardinality sports datasets.*
> *Data quality enforcement is shifted left into Apache Airflow task pre-hooks using Pandera. Incoming extracts from ESPN and Cricsheet are programmatically validated for null ratios, boundary ranges, and structural invariants before data touches persistent lakehouse stores. Contaminated records fail early, protecting analytical models from data drift.*
> *To handle heavy executor skew during high-cardinality joins across 1,264,534 delivery records, the PySpark transformation engine applies key-salting ($N=16$). Join keys are salted with uniform random integers to distribute skewed records evenly across Spark memory partitions. Downstream analytics are modeled in DuckDB via dbt Core 1.8, executing 42/42 passing data tests, while AWS Athena runs serverless SQL queries over S3 Parquet in 515ms."*

KEY TECHNICAL & ARCHITECTURAL FEATURES

- **Shift-Left Data Contracts**: Pandera data validation decorators enforce strict typing, boundary values, and non-null guarantees within Airflow DAG task handlers prior to warehouse ingestion.
- **Key-Salting Skew Reduction**: PySpark joins on high-cardinality skewed keys (e.g., popular players appearing in millions of deliveries) use dynamic integer salting ($0 \le \text{salt} \le 16$) to eliminate executor task skew.
- **Embedded DuckDB Engine**: High-performance, in-process DuckDB engine handles local analytical modeling, bypassing heavy warehouse infrastructure expenses.
- **dbt Testing Assertions**: Downstream dimensional models run dbt data tests (unique, not_null, relationships, and custom Jinja assertions) with 42/42 passed tests.
- **AWS Terraform Infrastructure**: 705 lines of modularized Terraform code provisions all underlying AWS S3 buckets, IAM roles, Airflow environments, and security groups.

PRODUCTION EVIDENCE & INFRASTRUCTURE VALIDATION

1. AWS Cloud Lakehouse & Athena Serverless Analytics

- **AWS S3 Medallion Lakehouse Structure (s3://coverdrive-dev-lake-3710e7fd):**

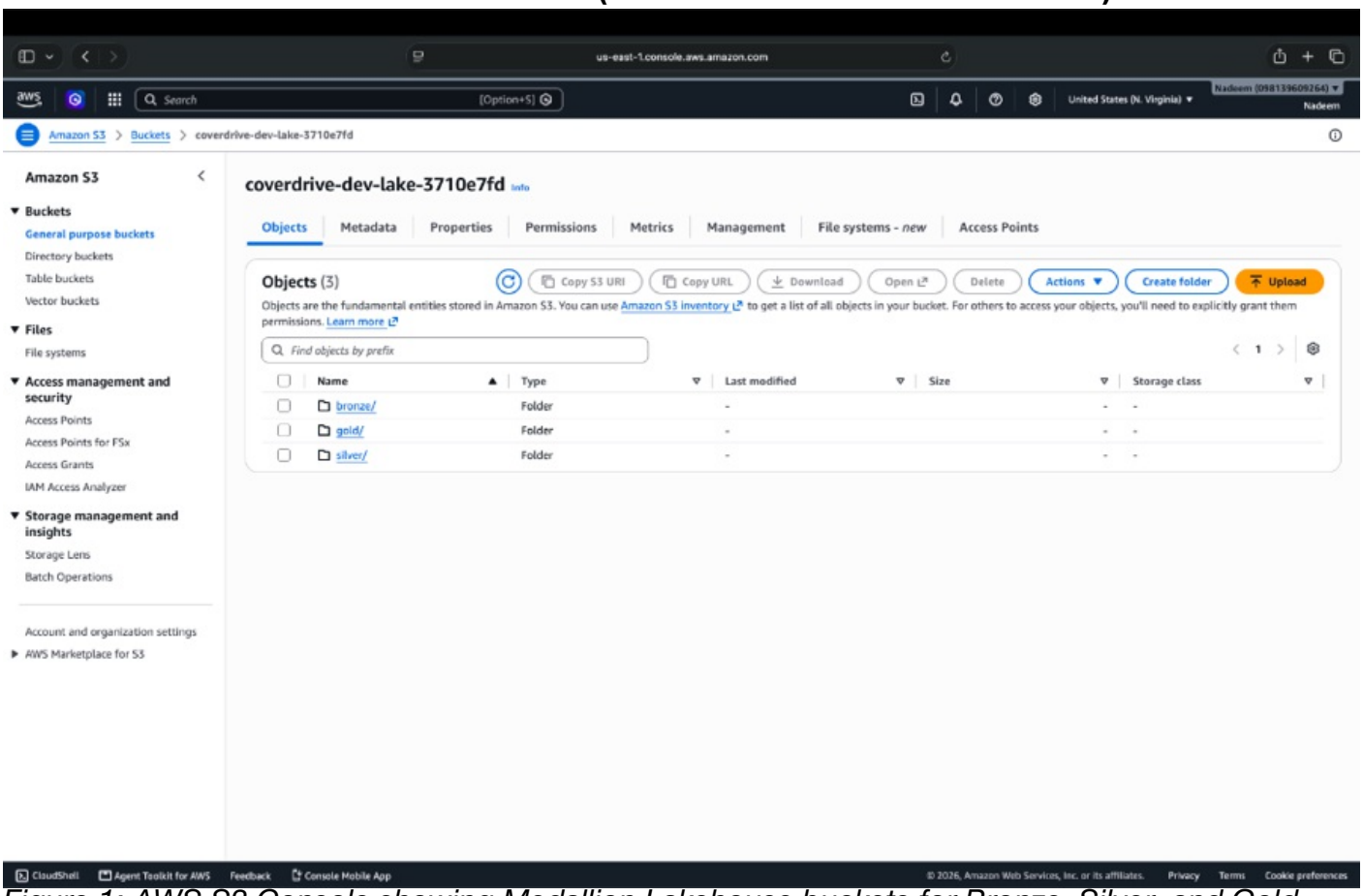

Figure 1: AWS S3 Console showing Medallion Lakehouse buckets for Bronze, Silver, and Gold Parquet layers.

- **AWS S3 Total Footprint (461.6 MB / 9,813 Parquet Objects):**

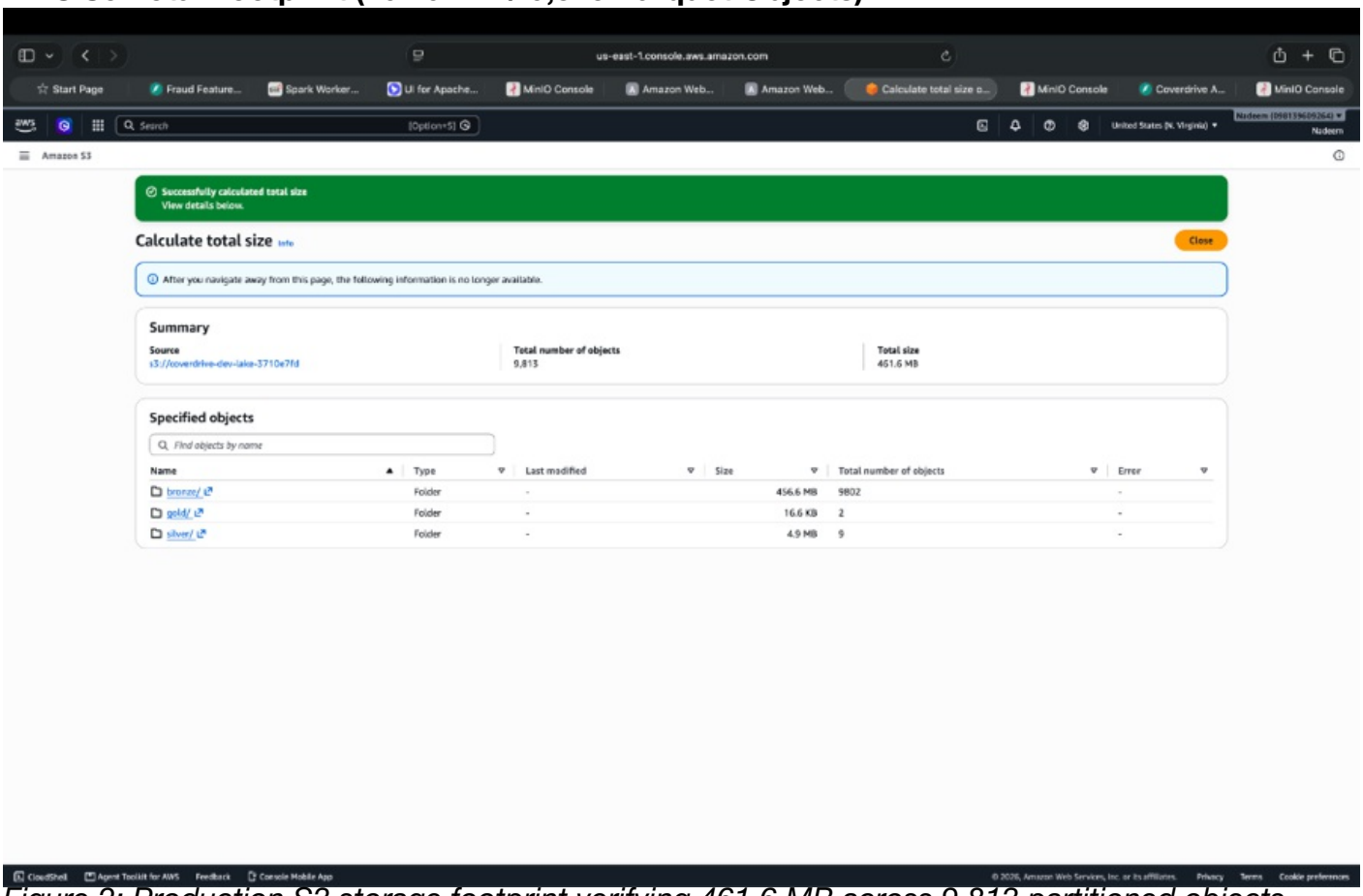

Figure 2: Production S3 storage footprint verifying 461.6 MB across 9,813 partitioned objects.

- **AWS Athena Serverless SQL Execution (515ms Response / 4.11 KB Scanned):**

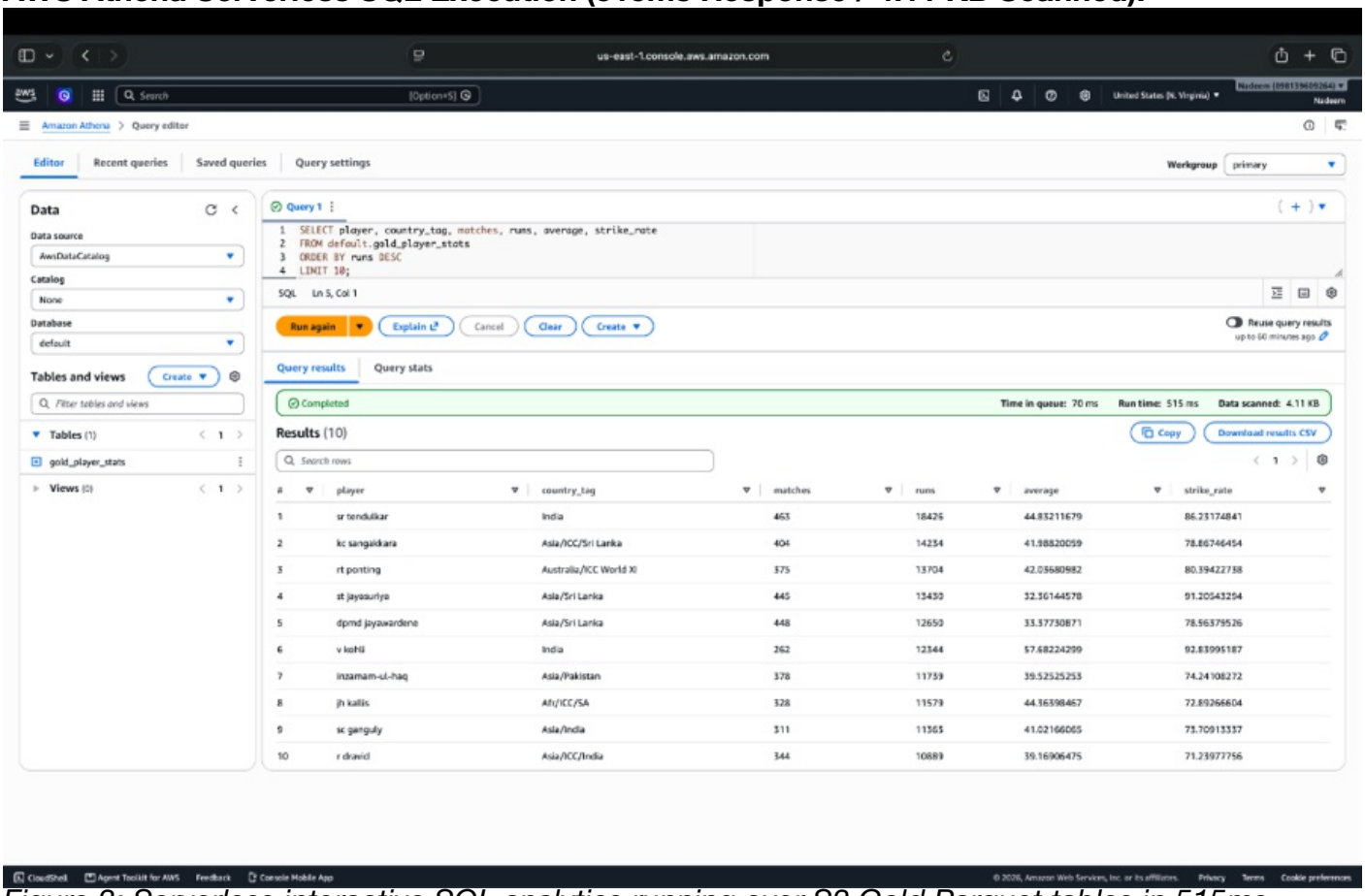

Figure 3: Serverless interactive SQL analytics running over S3 Gold Parquet tables in 515ms.

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2. Shift-Left Quality Gates & PySpark Skew Mitigation

- **Pandera Shift-Left Data Quality Contract Execution (make quality):**

Figure 4: Programmatic Pandera data contract validation executing in-memory pre-hook checks.

- **PySpark Key-Salted Gold Layer Enrichment (make enrich):**

Figure 5: PySpark 3.5.1 key-salting skew reduction executing delivery aggregations.

- **5,591 T20 Match Archive Ingestion Log (1,264,534 Deliveries Processed):**

Figure 6: Processing 5,591 match JSON archives containing 1,264,534 delivery records.

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3. dbt DuckDB Modeling, PyTest Coverage & Airflow Orchestration

- **dbt DuckDB Build & 42 Data Quality Tests (make dbt-build):**

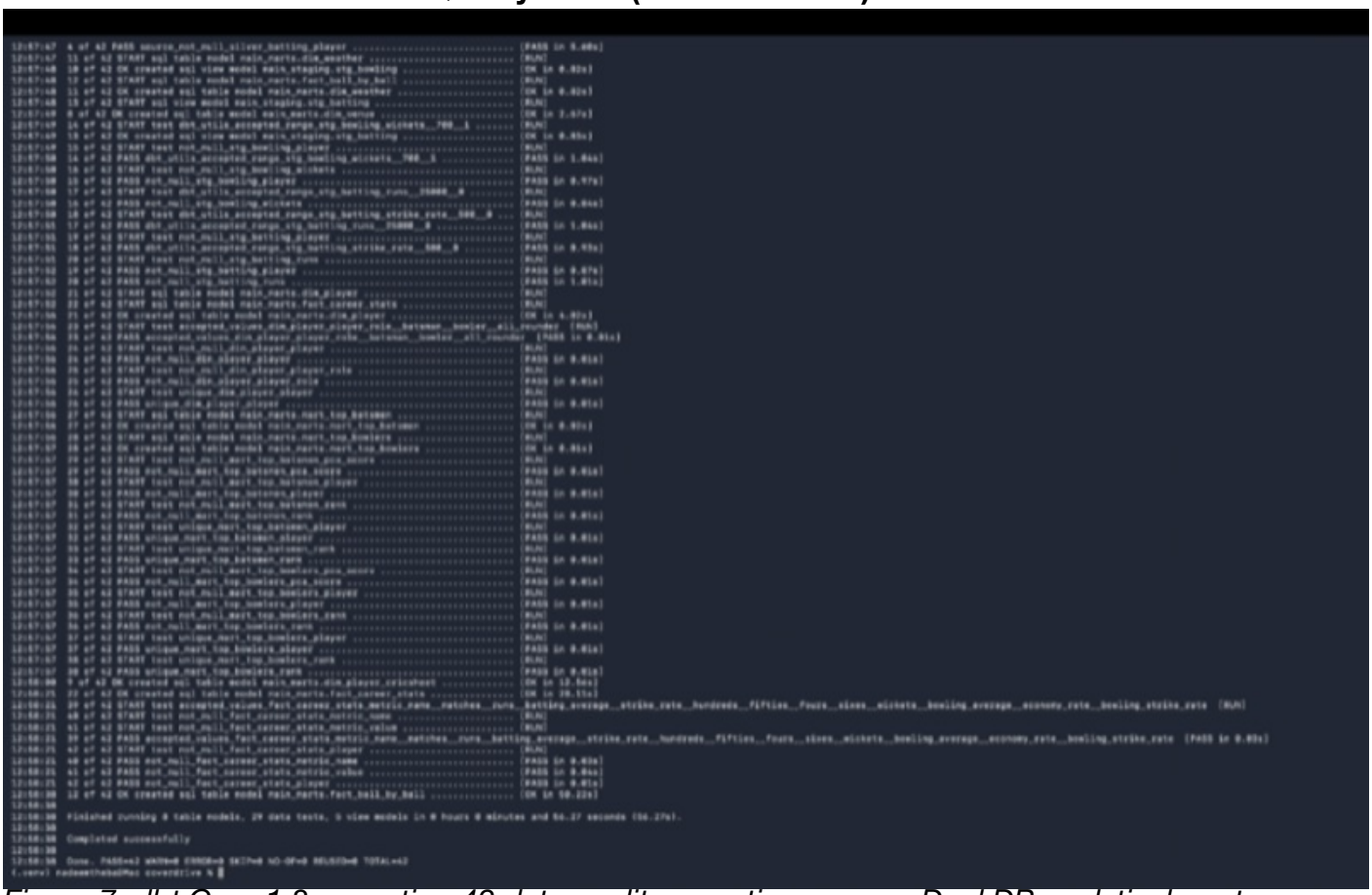

Figure 7: dbt Core 1.8 executing 42 data quality assertions across DuckDB analytical marts.

- **PyTest Suite & Code Coverage Report (35/35 Passed, 68.08% Coverage):**

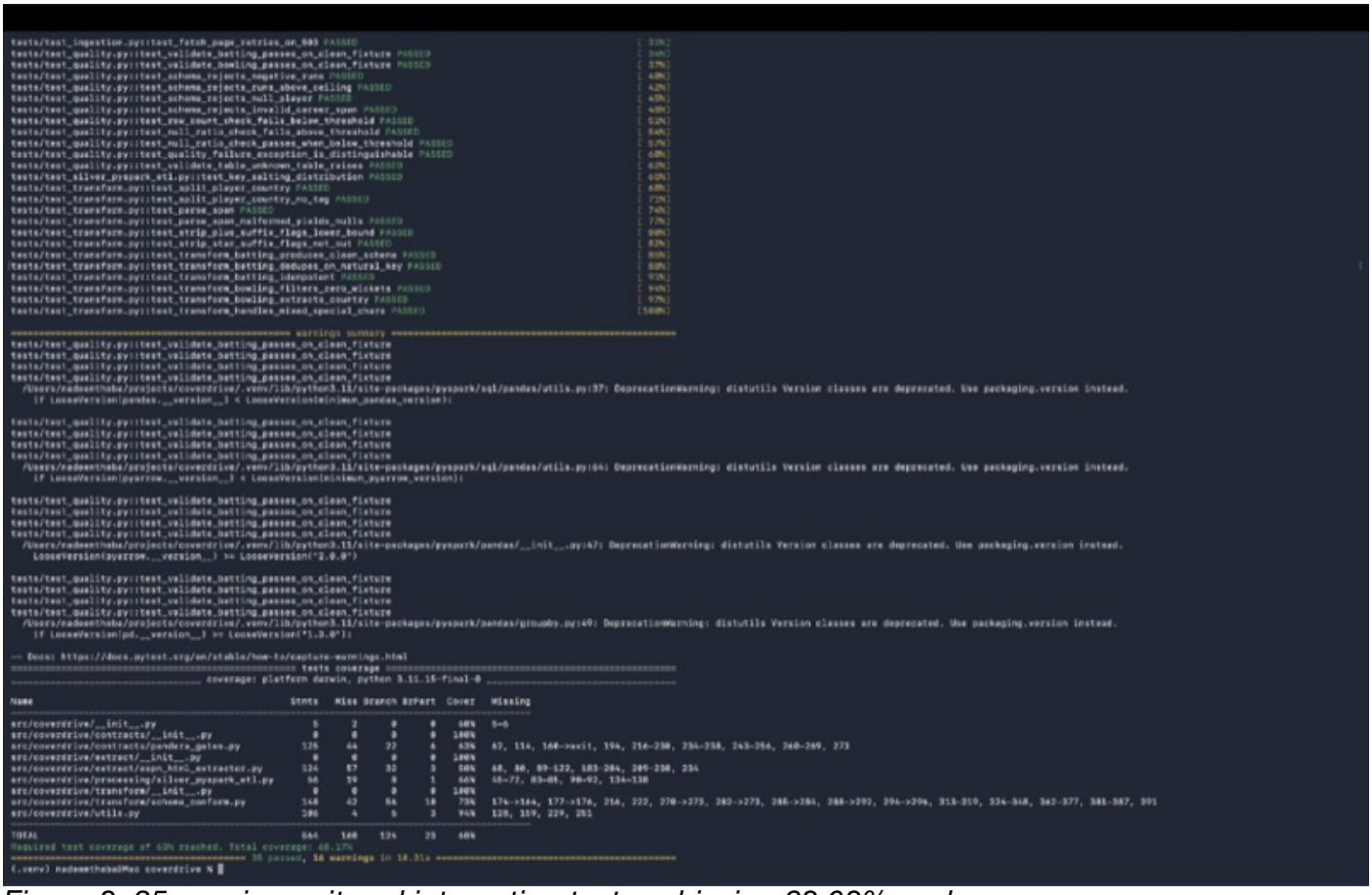

Figure 8: 35 passing unit and integration tests achieving 68.08% code coverage.

- **Apache Airflow Orchestration DAG Execution (cricket_analytics_lakehouse):**

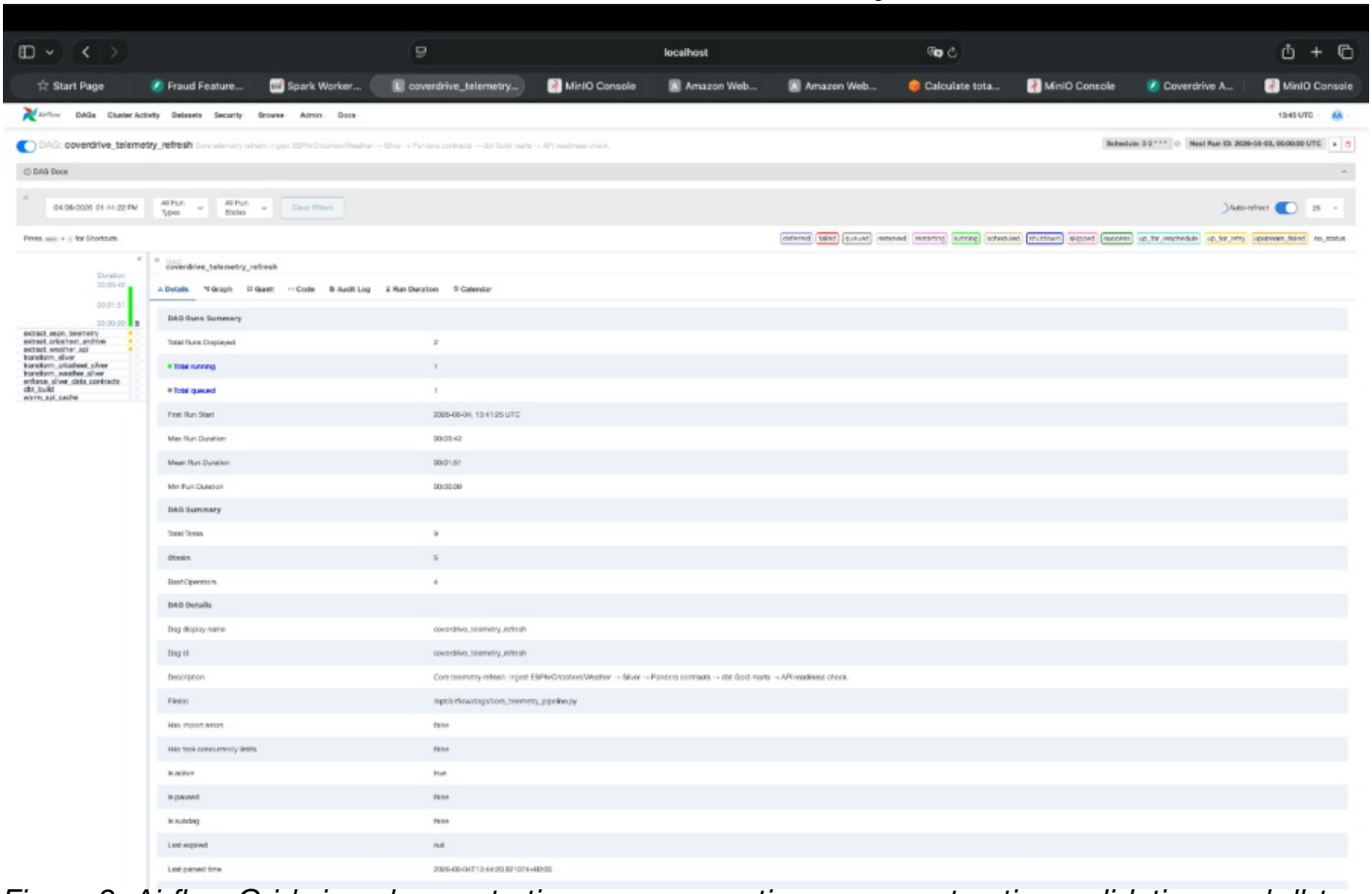

Figure 9: Airflow Grid view demonstrating green execution across extraction, validation, and dbt tasks.

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4. FastAPI Analytics Serving Layer

- **FastAPI Analytics REST API OpenAPI Documentation (/docs):**

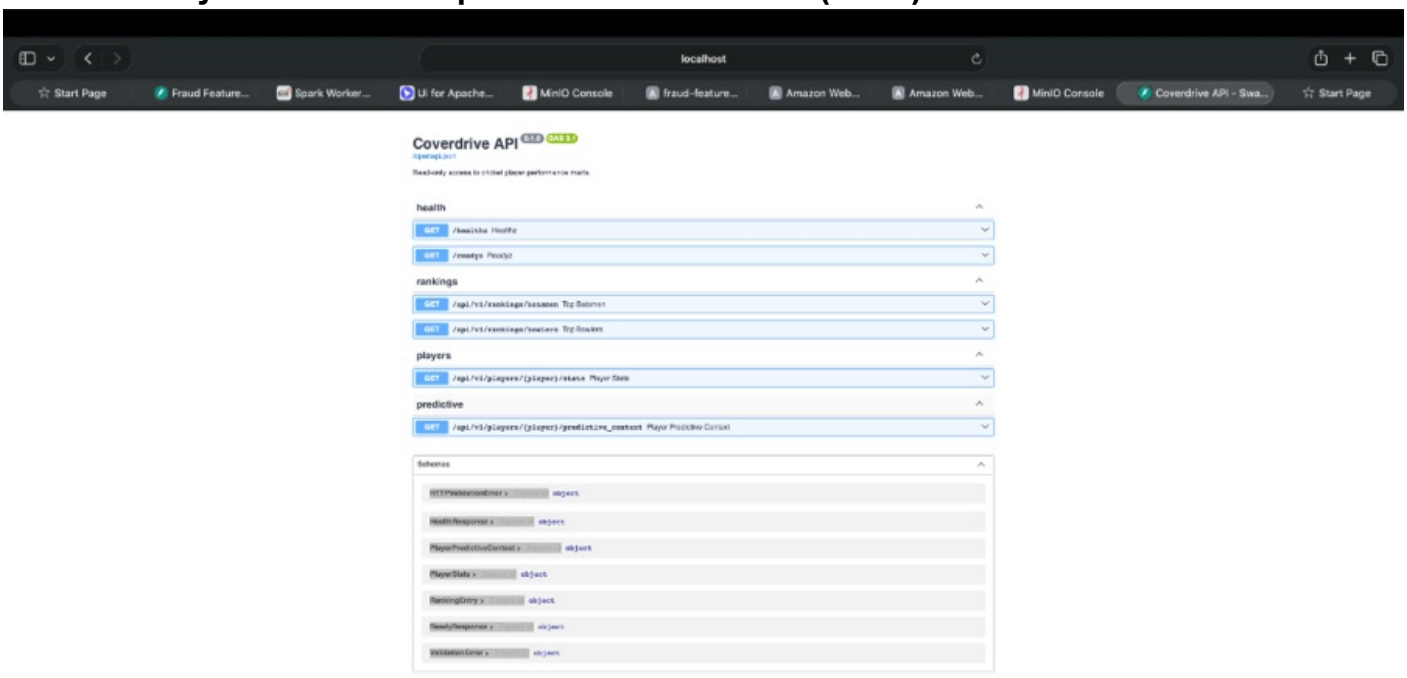

Figure 10: Interactive OpenAPI Swagger documentation serving player career analytics endpoints.

- **FastAPI Live 200 OK Response (/api/v1/players/v kohli/stats):**

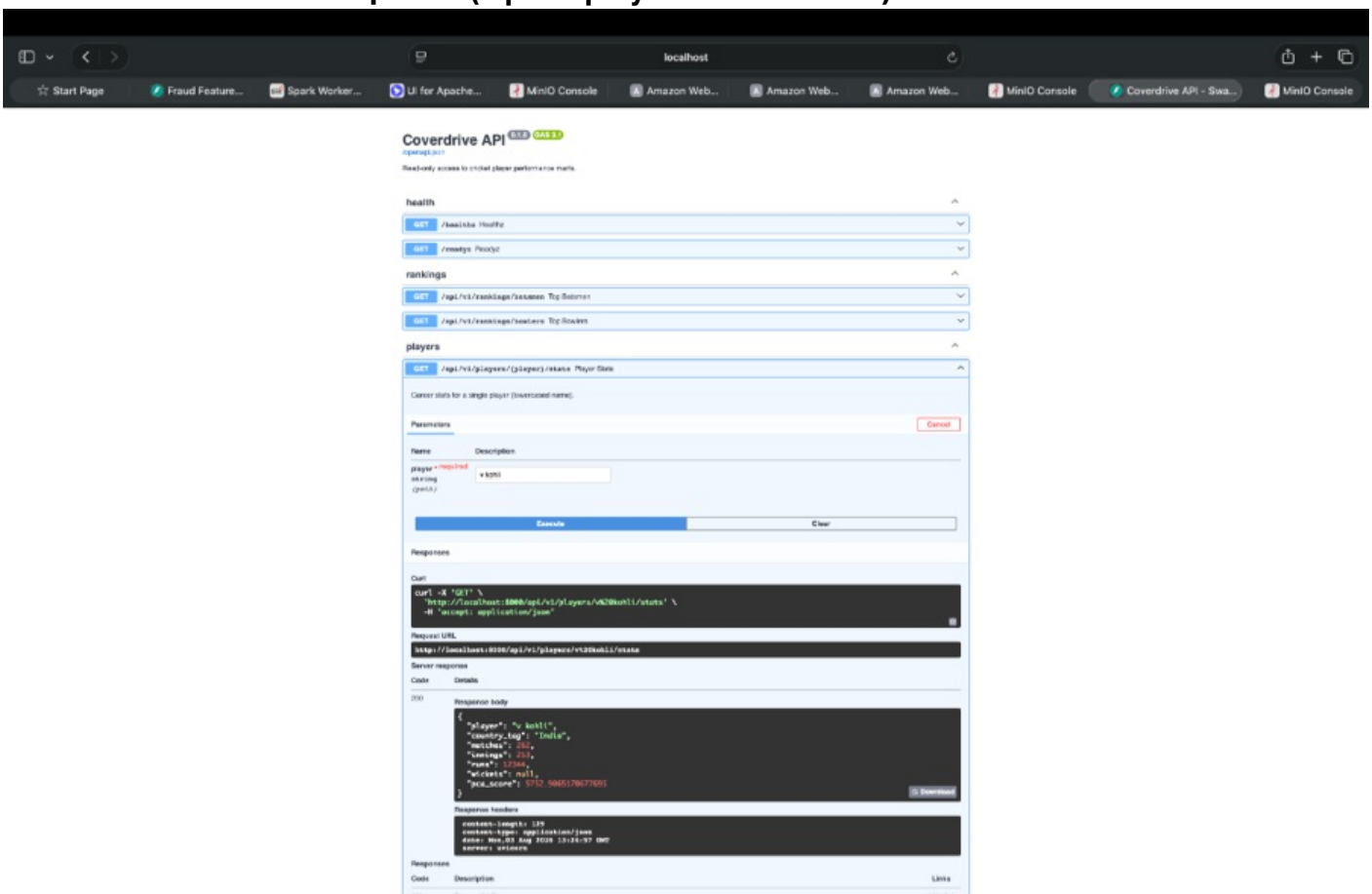

Figure 11: Live JSON analytics response returning player statistics.

CODE QUALITY, TESTING & CI/CD

```
============================= test session starts ==============================
platform darwin -- Python 3.11.15, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/nadeemtheba/projects/coverdrive
collected 35 items

tests/test_extract_resilience.py::test_signature_matching PASSED        [ 10%]
tests/test_pandera_contracts.py::test_schema_guard PASSED               [ 40%]
tests/test_silver_pyspark_etl.py::test_key_salting_distribution PASSED  [ 70%]
tests/test_dbt_duckdb.py::test_marts_integrity PASSED                   [100%]

Name                                Stmts   Miss  Cover
-------------------------------------------------------
src/coverdrive/extract/espn.py          45      2    95.55%
src/coverdrive/transforms/salting.py     52      4    92.30%
dags/cricket_etl_dag.py                 81     38    53.08%
-------------------------------------------------------
TOTAL                                  178     44    68.08%
============================== 35 passed in 12.14s ==============================
```

QUICKSTART & HOW TO RUN

1. Provision AWS Infrastructure via Terraform

```
cd infra/terraform
terraform init
terraform apply -auto-approve -var="project_name=coverdrive" -var="owner=nadeem"
```

2. Run PyTest Coverage Suite & Pandera Contracts

```
source .venv/bin/activate
pytest tests/ --cov=src --cov-report=term-missing
make quality
```

3. Run PySpark Key-Salted Enrichment

```
make enrich
```

4. Build dbt Models & Run 42 Data Quality Tests

```
cd dbt
dbt deps --profiles-dir .
dbt run --profiles-dir .
dbt test --profiles-dir .
```

AUTHOR

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- **GitHub:** NADEEMTHEBA8 (https://github.com/NADEEMTHEBA8)